

RESEARCH PROJECT

Prop-Fi Protocol: CDO-Based Real-World Asset Tokenization

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Prop-Fi Protocol: CDO-Based Real-World Asset Tokenization and Insured Liquidity Architecture

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Abstract

(This Abstract is written by focusing on "Expected Findings" as no simulation has been conducted.) Decentralized finance (DeFi) must solve the liquidity problem of the \$330 trillion [3] real estate market. This study presents an innovative protocol that directly addresses the operational and financial vulnerabilities [5] where CDOs failed during the 2008 Crisis. The protocol algorithmically manages the Coin supply by assessing credit risk with a two-dimensional dynamic matrix [6]. The most critical innovation is the Dual-Layer Insurance Mechanism [6], supported by a Pre-Liquidation Insurance Pool that instantly funds

the Bank's legal costs. Theoretical analyses and model comparisons are expected to show that the protocol, combined with the speed of the Avalanche network [7], can reduce legal foreclosure delay by over 80% and maintain its resilience against systemic shocks. The protocol offers an innovative theoretical model that brings modern solutions to modern problems.

Key Words: DeFi, RWA Tokenization, Real Estate Collateral, Hybrid DLT-RDBMS, CDO Risk Modeling, Dual-Layer Insurance, Algorithmic Lending, Smart Contracts, Risk Multiplier, Systemic Risk Management, Avax, Ethereum, Bitcoin

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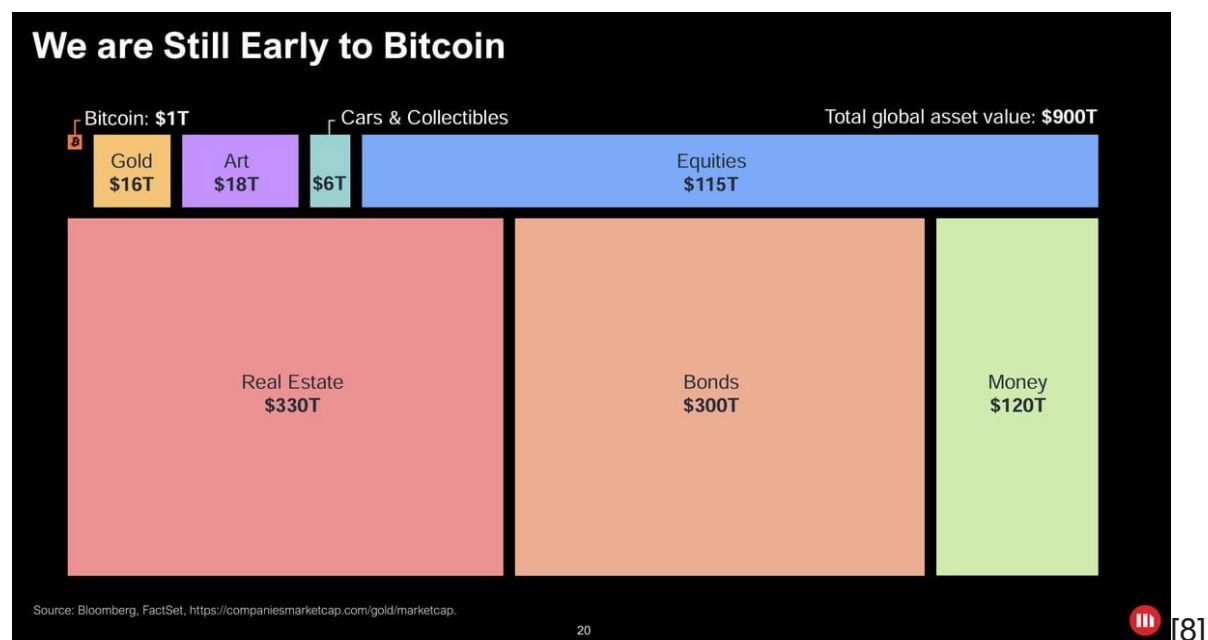
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1. Introduction

1.1. Global Financial Context and Problem Definition

The current DeFi ecosystem cannot capitalize on the liquidity of the \$330 trillion [3] real estate market. Analyses like 'Why We're Still Early...' [2], which demonstrate the crypto market's small share compared to total global wealth, reveal the necessity of integrating Real-World Assets (RWA). Traditional lending systems fail in the transition to decentralization due to legal enforcement delays and data reliability issues [4].



1.2. Research Aim and Original Contributions

This study aims to design a protocol that merges the strength of traditional finance (law) with the flexibility of DeFi (speed/transparency). The protocol's mechanism is based on the risk

tranching logic of Collateralized Debt Obligations (CDOs) [4], but aims to address the operational and financial vulnerabilities that emerged during the 2008 Global Financial Crisis [5].

Original Contributions: This protocol introduces the following key innovations [6]:

- **Hybrid Database Architecture:** Combines the immutability of DLT with the speed of RDBMS.
- **Dual-Layer Algorithmic Insurance:** Supported by a Pre-Liquidation Insurance Pool that instantly funds the Bank's legal costs.

2. Literature Review

2.1. Traditional CDO and 2008 Crisis Applications

The risk tranching logic of the protocol is based on the structure of Collateralized Debt Obligations (CDOs) [4]. CDOs collapsed in the 2008 Crisis due to Operational Vulnerability (delay in legal foreclosure) and Financial Vulnerability (static/faulty rating) [5]. Our protocol aims to directly address these two vulnerabilities within a decentralized architecture

2.2. Tokenization and Real-World Asset (RWA) Integration Challenges

This section examines the fundamental challenges in the RWA tokenization process (legal binding, property transfer, and valuation oracle issues) [4].

3. Methods

3.1. Hybrid Database Architecture and Data Integrity Innovations

The protocol combines the immutability of DLT with the speed of RDBMS [6].

A) Corporate Oracle Integration: Uses Banks as reliable data providers (Oracle) to feed personal credit history (R_K) data to the Smart Contract [6].

B) Feedback Loop (G_G): The Bank's legal foreclosure process delay data is fed into the algorithm; if there is a delay, the Bank's G_G multiplier decreases [6].

3.2. Dynamic Risk Rating and Algorithmic Multipliers

The Final Multiplier (F_{Rate}), which determines the quality of the Coin to be minted, is calculated with a two-dimensional matrix [6]:

$$F_{Rate} = (\alpha \times R_A) + (\beta \times R_K) - G_G \quad [6]$$

- α and β (Weighting Coefficients): Show the importance the protocol gives to the quality of the property (α) and the behavior of the borrower (β) when pricing the risk ($\alpha + \beta = 1$).

3.3. Algorithmic Coin Supply and Mathematical Model

CDO-Crypto Multiplier (Tranching Analogy): The protocol algorithmically tranches the lending amount using the property collateral quality (R_A multiplier) [6].

Core Lending Operation (Minting):

$$A_{Yeni} = V_{M\ddot{u}lk} \times F_{Rate} \times LTV_{Maks} \quad [6]$$

Financial Insurance: Coin Burning: When the loan is repaid, the principal part of the Coin is automatically burned [6].

3.4. Dual-Layer Insurance and Decentralized Liquidation Mechanism

Pre-Liquidation Insurance Pool (Operational Insurance): Instantly funds the Bank's legal costs with funds staked by those who wish to participate in the auction [6].

Participation Right: Those who stake this fund gain the right to participate in the decentralized auction [6].

3.5. Role and Applications of Smart Contracts

Smart Contracts perform the calculation of the Final Rate (F_{Rate}), Coin minting, management of the Fund Insurance Pool, and the automatic adjustment of the G_G multiplier [6].

4. Experiments

(Experiments will be conducted using Simulation / Benchmarking methods on 1,000 synthetic portfolios.) [6]

Exp. No	Focus Area	Scenario Description	Key Performance Indicator (KPI)
Exp 1	Hybrid Data Speed	Time required to pull data from the Bank Oracle and assign the risk multiplier to the property.	Oracle Latency and Total Loan Approval Time [7].
Exp 2	Algorithmic Risk Control	Comparison of the Coin amounts minted in response to different α and β values.	CC Mint Reduction Rate (Percentage reduction in the minted Coin amount compared to $\alpha\alpha$).
Exp 3	Systemic Crash Stress	A crisis simulation (7 days) where all Real Estate Values drop by 30% and the Coin price deviates by 50% from its intrinsic value.	Pool Depletion Time and Coin Burning Effectiveness (How many days it takes to correct the price deviation).

5. Results & Discussion

5.1. Key Arguments and Expected Legal Financing Impact

The protocol is expected to significantly reduce the legal foreclosure delay by instantly financing legal costs in a decentralized manner, thanks to the Pre-Liquidation Insurance Pool [6].

5.2. Systemic Risk and Shock Resilience

The stability of the real asset collateral [4] protects the protocol from systemic shocks. Experiment 5 will theoretically prove the effectiveness of the Financial Insurance [6].

5.3. Decentralization Trade-off and Bank Risk Management

The Feedback Loop (G_G multiplier) mechanism manages risk by enforcing the honesty of the central actor, the Bank [6].

6. Conclusion and Future Work

6.1. Research Contributions and Financial Revolution Perspective

The core success of our protocol is designing mechanisms that technically address the two main vulnerabilities where CDOs failed in the 2008 Crisis [5].

Summary of Technical Contributions: With architectural and algorithmic innovations such as the Hybrid Database Architecture, Dual-Layer Algorithmic Insurance [6], and the Feedback Loop, the system offers a unique data management mechanism that enforces the honesty of the central actor.

According to Deloitte's reports [1], the total value of the tokenized real estate market on the blockchain was below \$300 billion in 2024; however, this value is projected to exceed \$4 trillion by 2035. Our protocol facilitates the transition for this market with huge potential.

6.2. Future Work and Recommendations

(This section will include future research topics for your paper (e.g., Automatic update of the R_A score with machine learning, cross-chain integration).)

Future work must focus on **rigorous statistical testing** of the Final Risk Rate formula (F_{Rate}) to **optimize weighting coefficients** (α, β) across various economic conditions. This methodological advancement will be supported by the integration of **Machine Learning (ML)** for dynamic, predictive risk scoring of assets (R_A), moving beyond static appraisals to real-time, data-driven valuations.

The ultimate strategic objective is to achieve **cross-chain interoperability** with major DLT ecosystems like Ethereum, expanding the protocol's market reach and liquidity. Furthermore, minimizing reliance on centralized corporate actors (Banks) will require the development and integration of **Decentralized Legal Oracles** to provide tamper-proof, real-world legal data on foreclosure status directly to the Smart Contracts.

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